

Automated Review Rating System

1 Project Overview

An intelligent system that analyzes customer reviews and predicts corresponding star ratings. It uses Natural Language Processing (NLP) to extract sentiment and key features from textual feedback. The model is trained on labeled review data to learn patterns between language and rating. This enables automated, consistent, and scalable review evaluation, helping businesses gain actionable insights and improve decision-making.

2 Environment Setup

- Python 3.9+ was used as the base language.
- Key libraries: Pandas, NumPy, Scikit-learn, Joblib, Matplotlib, Seaborn.
- IDE Used Jupyter Notebook
- Verified proper functioning of packages for data preprocessing, NLP, and data visualization.

3 GitHub Project Setup

The screenshot displays a GitHub repository interface for the user 'Akash-VS2002'. At the top, navigation elements include the 'main' branch selector, '1 Branch' and '0 Tags' indicators, a search bar labeled 'Go to file', an 'Add file' button, and a green 'Code' button. Below the repository header, a table lists the project's files and folders with their commit messages and timestamps.

File/Folder	Commit Message	Time
app	Restore deleted app/kaggle folder	last week
data	Add files via upload	now
frontend	Delete frontend/kaggle directory	last week
models	Add folder structure with README and requirements	last week
notebooks	Add folder structure with README and requirements	last week
README.md	Initial project structure with README	last week
requirements.txt	Initial project structure with README	last week

Below the file list, the 'README' file is selected and its content is displayed. The title 'Automated Review Rating System' is prominently shown, followed by a descriptive paragraph about the project's functionality using NLP and Machine Learning.

Automated Review Rating System

This project is an Automated Review Rating System that uses Natural Language Processing (NLP) and Machine Learning to analyze customer reviews and generate ratings automatically.

4 Data Collection

Datasets Collected from Kaggle:

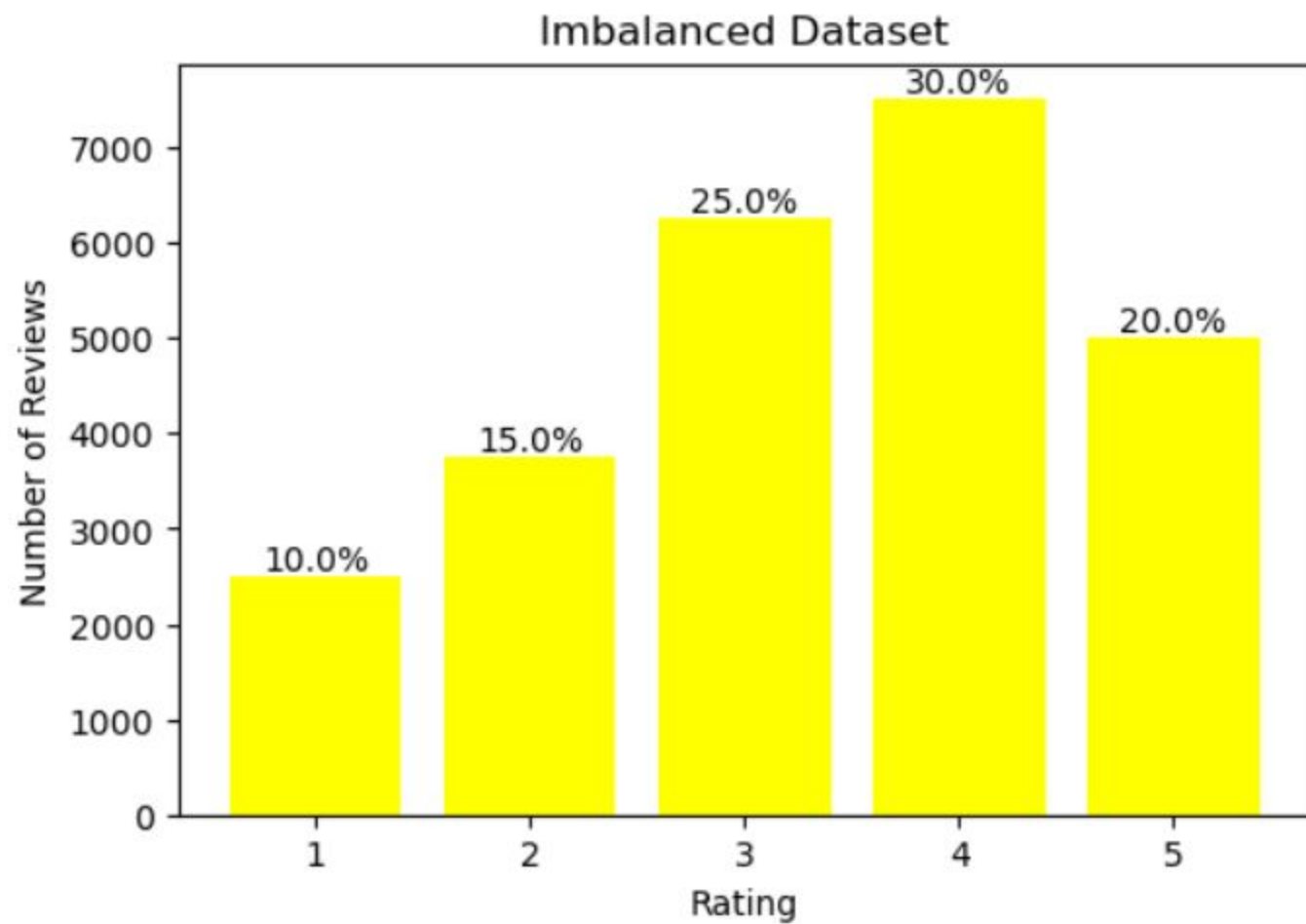
- <https://www.kaggle.com/datasets/snap/amazon-fine-food-reviews>
 - Data includes user reviews and corresponding ratings (1–5 stars)
 - Ensured Imbalanced representation of ratings for better model training

4.1 Imbalancing the Dataset

The dataset contains **25,000 customer reviews**, distributed across five star ratings in an **imbalanced manner**. Unlike a balanced dataset, the reviews are unevenly spread:

- **1★**: 10% (2,500 samples)
- **2★**: 15% (3,750 samples)
- **3★**: 25% (6,250 samples)
- **4★**: 30% (7,500 samples)
- **5★**: 20% (5,000 samples)

imbalanced dataset 🖱️ [Imbalanced_Dataset.csv](#)



5 Data Preprocessing

Data preprocessing involves cleaning, transforming, and organizing raw data to make it suitable for analysis or modeling. It ensures **accuracy, consistency, and better model performance** in machine learning tasks.

5.1 Removing Duplicates

Identify and remove duplicate reviews to avoid bias and redundancy in the dataset.

5.2 Detect and Remove Confusing Reviews

Detect and remove reviews where the text does not match the given star rating, ensuring label accuracy.

5.3 Dropping Unnecessary Columns

Remove columns from the dataset that do not contribute to review rating prediction. Reduces noise and improves model training efficiency.

Code 👉 `df = df.drop(columns=['review_id', 'username', 'date'])`

5.4 Handling Missing Values

Identify and handle missing or null values in the dataset to ensure clean and complete data. Missing values in reviews or ratings can negatively affect model training and predictions.

Code 👉 `df = df.dropna()`

5.5 Converting Text to Lowercase

Convert all review text to lowercase to maintain uniformity. Helps the model treat words like **Good** and **good** as the same token, reducing redundancy.

Code 👉 `df['review_text'] = df['review_text'].str.lower()`

5.6 Removing URLs, Emojis, and Special Characters

Clean review text by removing unnecessary elements like URLs, emojis, and special characters.

Helps reduce noise and ensures that the text is suitable for feature extraction and modeling.

Code 👉 `text = re.sub(r'http\S+|www.\S+', "", text), text = text.encode('ascii', 'ignore').decode('ascii'), text = re.sub(r'^a-zA-Z0-9\s', "", text)`

5.7 Removing Punctuations

Remove punctuation marks (like . , ! ? ; :) from review text. Helps reduce noise and ensures that words are treated consistently during feature extraction.

Code 👉 `df['review_text'] = df['review_text'].str.replace(f"[{string.punctuation}]", "", regex=True)`

5.8 Stopword Removal

Stopwords are common words (e.g., “the”, “is”, “and”) that do not contribute significant meaning to text analysis.

Removing them reduces noise and improves model performance for sentiment or review rating prediction. Total stopwords removed: 106161

Removed stopwords

['a', 'about', 'above', 'after', 'again', 'against', 'all', 'am', 'an', 'and', 'any', 'are', 'as', 'at', 'be', 'because', 'been', 'before', 'being', 'below', 'between', 'both', 'but', 'by', 'can', 'd', 'did', 'do', 'does', 'doesn', 'doing', 'don', 'down', 'during', 'each', 'few', 'for', 'from', 'further', 'had', 'has', 'have', 'having', 'he', 'her', 'here', 'hers', 'herself', 'him', 'himself', 'his', 'how', 'i', 'if', 'in', 'into', 'is', 'it', 'its', 'itself', 'just', 'll', 'm', 'ma', 'me', 'more', 'most', 'my', 'myself', 'no', 'nor', 'not', 'now', 'o', 'of', 'off', 'on', 'once', 'only', 'or', 'other', 'our', 'ours', 'ourselves', 'out', 'over', 'own', 're', 's', 'same', 'shan', 'she', 'should', 'so', 'some', 'such', 't', 'than', 'that', 'the', 'their', 'theirs', 'them', 'themselves', 'then', 'there', 'these', 'they', 'this', 'those', 'through', 'to', 'too', 'under', 'until', 'up', 've', 'very', 'was', 'we', 'were', 'what', 'when', 'where', 'which', 'while', 'who', 'whom', 'why', 'will', 'with', 'won', 'y', 'you', 'your', 'yours', 'yourself', 'yourselves']

Code 🙌

```
def remove_stopwords(text):  
    text = str(text).lower()  
    text = re.sub(r'^a-z\s]', "", text)  
    words = text.split()  
    cleaned_words = [lemmatizer.lemmatize(w) for w in words if w not in stop_words]  
    return " ".join(cleaned_words)  
  
df = pd.read_csv(r"C:/Users/MY PC/OneDrive/Desktop/kaggle/Imbalanced_Dataset.csv")  
df['Cleaned_Review'] = df['Review'].apply(remove_stopwords)
```

5.9 Lemmatization

Lemmatization is the process of reducing words to their dictionary or base form (lemma). It considers the context and meaning of the word. Unlike simple stemming, lemmatization produces real words that exist in the language. This helps in preserving the semantic meaning of the text, which improves the accuracy of NLP tasks like sentiment analysis, text classification, and review rating prediction.

Example:

- Words like **running**, **ran**, **runs** → lemmatized to **run**
- Words like **better** → lemmatized to **good**

Why Lemmatization is Better than Stemming

meaning Preservation:

- Lemmatization reduces words to their **real dictionary form (lemma)**, preserving the meaning.
- Stemming simply cuts word endings, which can produce non-words or incorrect forms.

Accuracy:

- Lemmatization considers the **context and part of speech**, making it more accurate.
- Stemming uses a heuristic approach, often leading to errors.

Example:

- Lemmatization: **running** → **run**, **better** → **good**
- Stemming: **running** → **run**, **better** → **better** (unchanged, may be incorrect)

5.10 Filter by Word Count

Filtering reviews by word count ensures that extremely short or empty reviews, which may not provide meaningful information, are removed. This improves model training by focusing on reviews with enough content to learn patterns.

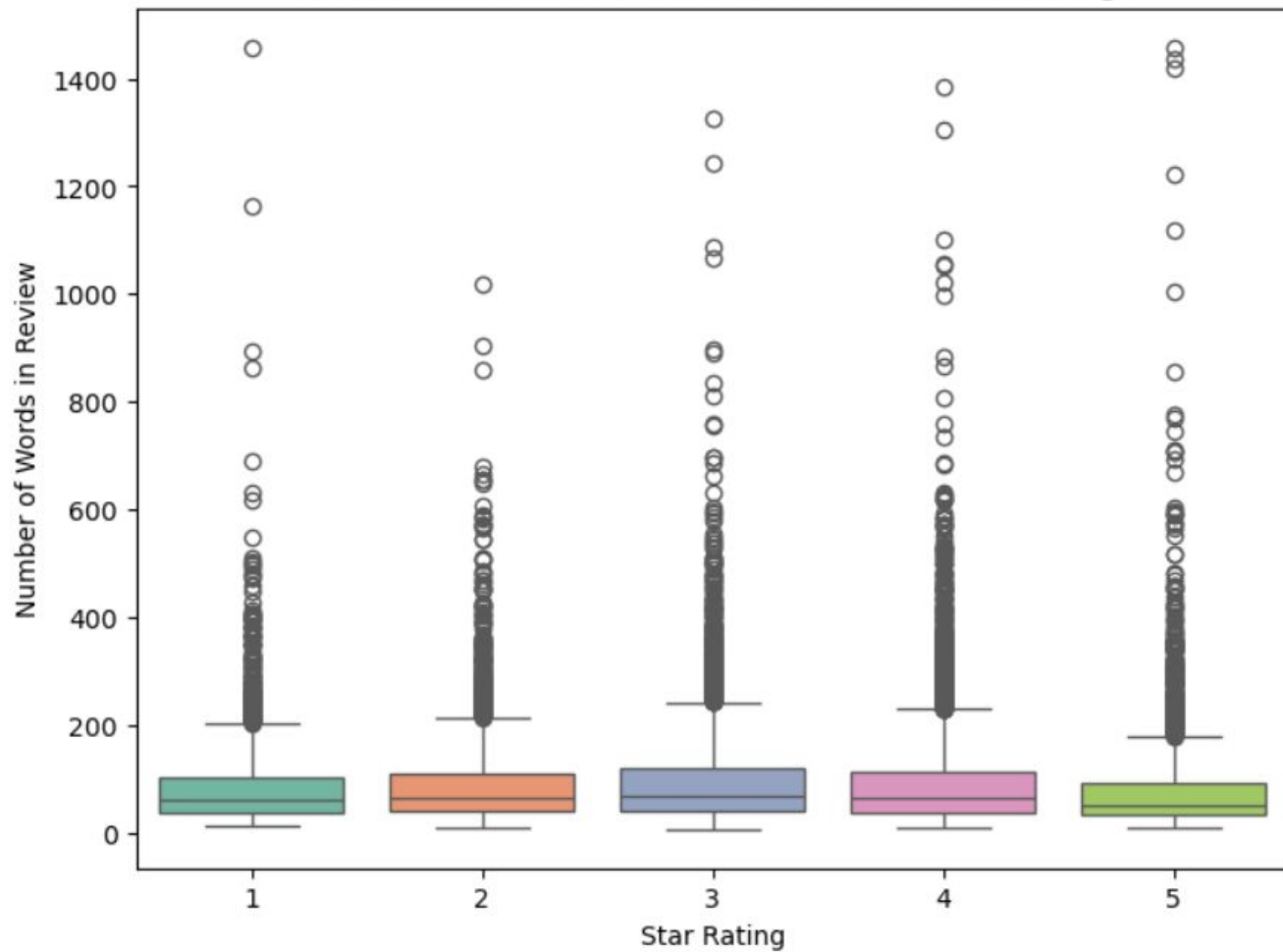
6 Data Visualization

Data visualization helps to **understand the dataset**, identify patterns, and communicate insights effectively before model training. In the Automated Review Rating System, we visualize reviews, ratings, and text characteristics.

Box Plot

A **box plot** is a visual tool used to display the distribution of numerical data, showing the **median, quartiles, and potential outliers**. It helps quickly understand how data is spread and identify any extreme values. In this project, **box plots were used to compare the word count distribution of reviews across different star rating classes**, allowing us to analyze patterns such as whether longer reviews tend to correspond to higher or lower ratings.

Distribution of Review Word Count Across Star Ratings



Sample of 1-Star Rating

1. dont know received burger colorless mine red wonder safe eat
2. ive amazon since ive ordered item first review ive written read review item figured seller would learn try put many different sampler item opened package total kcups horrible pumpkin spice flavor couldnt get rid save life extremely upset feel ripped avoid seller cost got much better deal shoffeecom
3. product taste like expected sampling plum sauce like one would purchase item
4. last year another turducken great year market brought never buy one instruction state cook hour minute frozen took hour could lived took dry outside still raw middle trust market heard several complaint yearbr pa
5. received today wife horrible tasting coffee ever wasnt case caring coffee right nasty wife thought dont know got bad batch wont ordering

Sample of 2-Star Rating

1. nice convenient package hummus cracker good combo however content package gluten free chocolate nut mix produced equipment also process wheat celiac disease sensitive gluten two item completely gluten free
2. use whole lot little vanilla flavor good value vanilla extract work better everytimebr bottle beautiful practical wire top hard move order open bottle also shaker spoon metal cap hole held plastic cap underneath doesnt hole whats deal
3. excited try item calorie well sound good trueit doesnt taste anything like peanut whatsoever really wanted like unfortunatley wasted money dont waste
4. bought gum remember chewing kid thought juciy filling bestbr well see cant found store anymore hard liquid center like gel lost flavor within five minute really upset stuck case waste money
5. ok im deployed thought would buy something odd far beef jerky guy troop person liked jerky bad good

Sample of 3-Star Rating

1. disappointed product chewed gum get rubbery fast wouldnt recommend anyone else
2. make great waffle contrary description need add oil egg mix rather water probably issue people someone temporarily without kitchen fridge like u little disappointing
3. received order french lentil barry farm direction cooking lentil additional research know use something excited
4. kind healthy grain vanilla blueberry cluster flax seed better snack eat cereal milk taste acquired taste youve eaten start taste better never taste bad taste little like cardboard get used id recommend people health conscience like food like isnt something buy
5. dont know java one theyre managing put good coffee defective pod last shipment least one five pod sealedyou open envelope loose coffee edge pod open due opening incorrectly filter paper ripped damaged isnt glued sealed properly definitely recommended buy unless want waste money like coffee stay pod

Sample of 4-Star Rating

1. enjoyed soup yes would like fresh pho soup area live find vietnamese restaurant mile made soup added fresh bean sprout fresh cilantro fresh green onion strip cooked steak basic soup add thing itnot expect open strip steak going appear box recommend soup starter great soup adding fresh ingredient fresh thing never let
2. make week special treat there toasted coconut coconut powder crossed mind buy aunt jemima sp add shredded coconut tho sure would good probably worth try since canister pricey tho pricey bought bulk
3. way spicy love amys product feel like warning heat index
4. looking cereal white flour white sugar cereal taste great fill
5. nice bubbly beverage didnt satisfy desire soda lol yummy though

Sample of 5-Star Rating

1. bought lb merckens coco lite coating make chocolate covered strawberry valentine day tasted great melted well dried quickly strawberry great
2. began ordering using subscribe save program posted overseas back state probably continue buy amazon price way better grocery store around place compete walmart bit hike literally son love cereal mysteriously disappears much rapidly weekend husband eats breakfast home although healthier cereal happy one made whole grain relatively low sugar content especially compared sugary cereal available overseas believe u thanks amazon carrying product
3. fantastic people cant eat sugar ie diabetic youve scoured globe looking sugarfree snack doesnt fake sugar flavour search endedbr br huge haribo fan widely available u friend germany send haribo candy time biggest fear would live expectation relieved find barely tell differencebr br one word caution mentioned several reviewer cannot eat large quantity go sugar free candy stomach discomfort flatulencegas diarrhea wellknown side effect sugarfree candy wise eat dozen piece
4. husband lived spain year missing yummy food ate ordered cheese blown away great tasted husband loved real spanish cheese well ordering assortment
5. skeptical first actually feeling lot better finger joint also nail became healthy strong keeping french natural nail mixing soy protein shake morning breakfast doesnt strange taste sonperfect

7 Train–Test Split

The dataset is divided into two parts:

- **Training set (80%)** → used to train the model.
- **Testing set (20%)** → used to evaluate model performance on unseen data.

Stratified sampling is applied to ensure that all star rating classes (1★–5★) are proportionally represented in both training and testing sets, according to their **imbalanced distribution**:

- 1★ → 10%
- 2★ → 15%
- 3★ → 25%
- 4★ → 30%
- 5★ → 20%

This approach prevents the model from being biased toward any single rating class while still reflecting the real-world imbalanced distribution of reviews.

How It Was Done

The dataset contains **25,000 reviews** with an **imbalanced distribution** across five rating classes: 1★ (10%), 2★ (15%), 3★ (25%), 4★ (30%), and 5★ (20%). For effective model training and evaluation, the dataset was split into **20,000 training samples (80%)** and **5,000 testing samples (20%)**.

The `train_test_split` function was applied with the **stratify parameter**, ensuring that the imbalanced class proportions were preserved in both the training and testing sets. This maintained the natural imbalance of the dataset while preventing the model from being biased toward any particular rating class.

Training Dataset: [imbalanced_train.csv](#)

Testing Dataset: [imbalanced_test.csv](#)

Code 🙌 `df.columns = df.columns.str.strip()`

`# Features and target`

`X = df['Review']`

`y = df['Rating']`

`# Stratified train-test split (80%-20%)`

`X_train_text, X_test_text, y_train, y_test = train_test_split(`

`X, y, test_size=0.2, random_state=42, stratify=y`

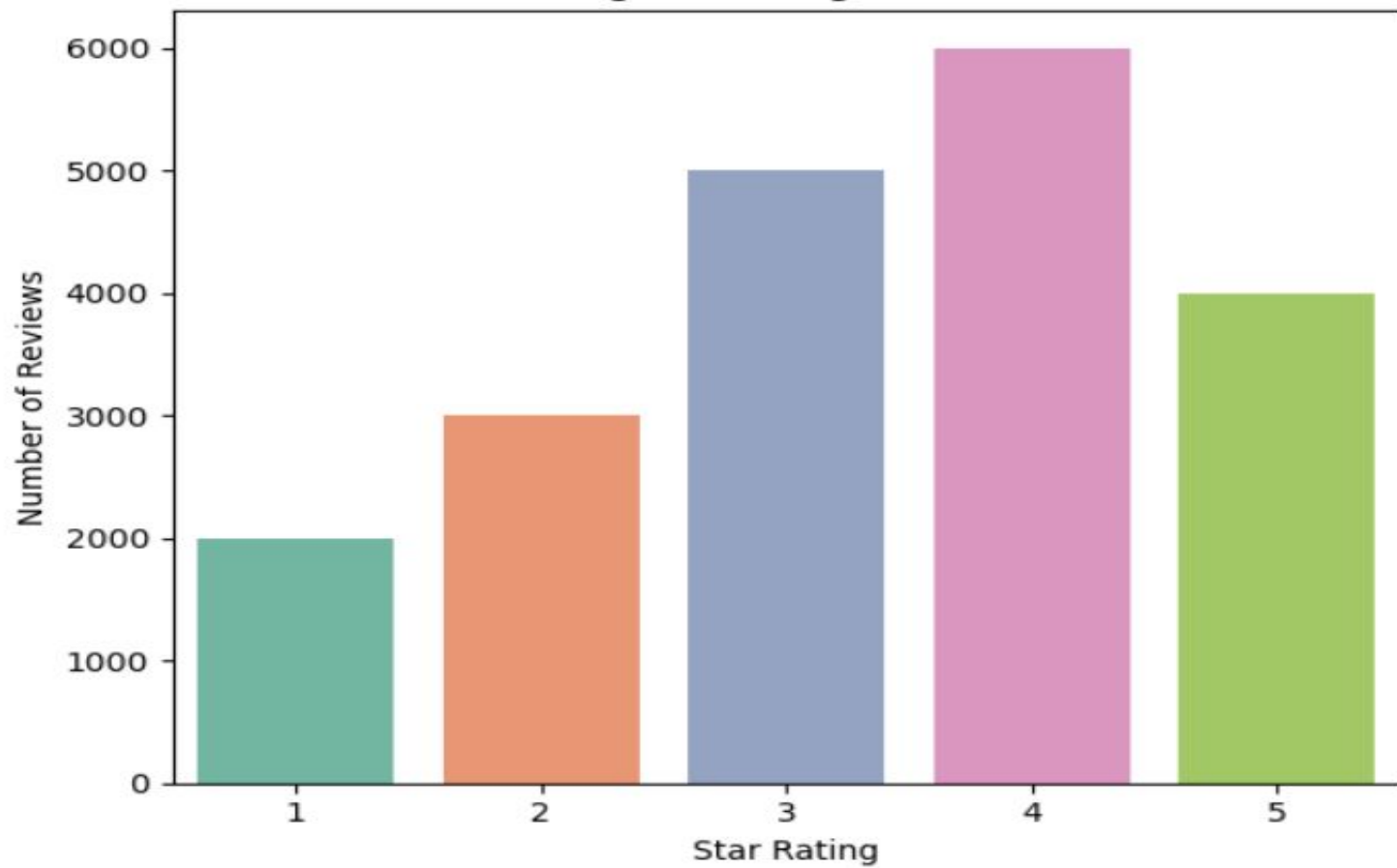
`)`

`# Combine features and target for saving`

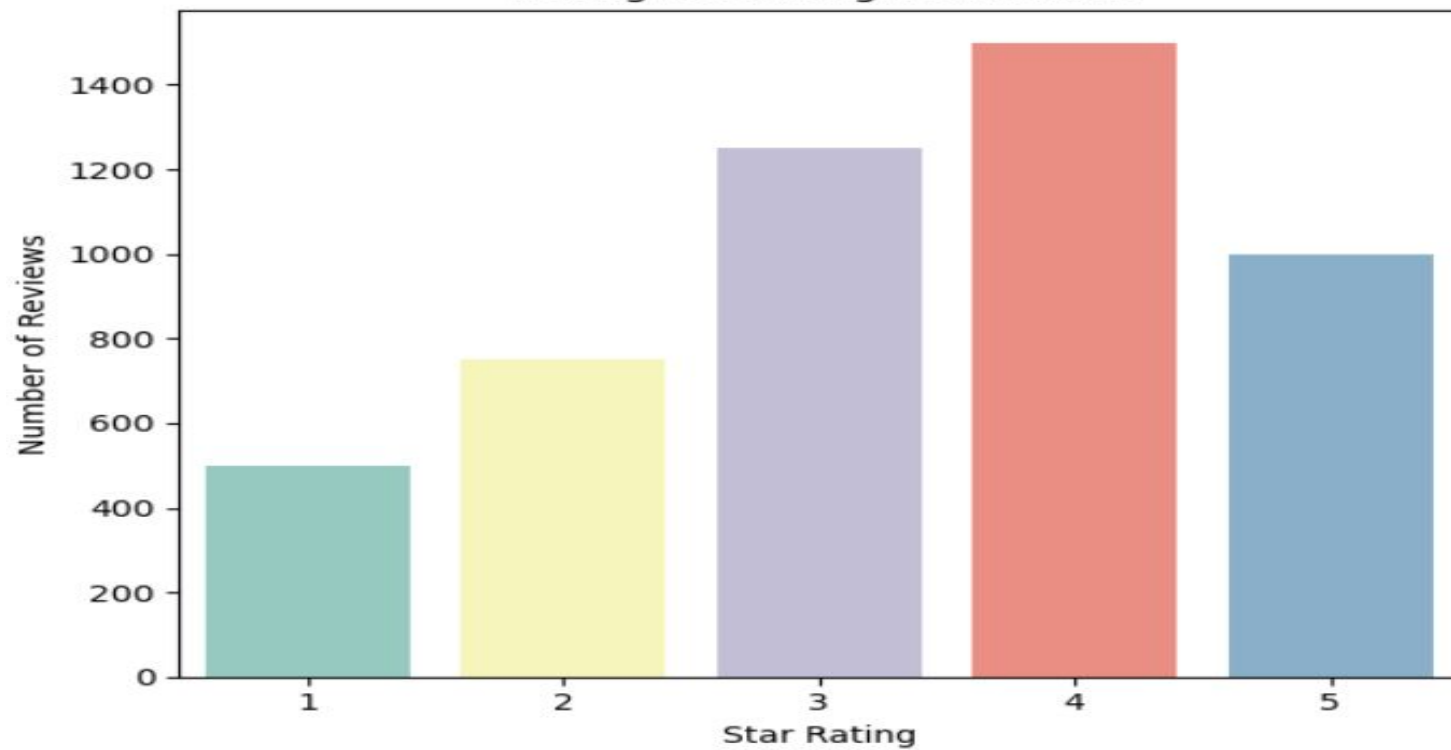
`train_df = pd.DataFrame({'Review': X_train_text, 'Rating': y_train})`

`test_df = pd.DataFrame({'Review': X_test_text, 'Rating': y_test})`

Training Set Rating Distribution



Testing Set Rating Distribution



8 Text Vectorization

Text vectorization transforms textual reviews into numerical representations that machine learning models can process. It converts words into features, capturing patterns and relationships within the text for effective model training.

TF-IDF (Term Frequency–Inverse Document Frequency)

TF-IDF is a text vectorization technique that converts words into numerical values by combining **Term Frequency (TF)** and **Inverse Document Frequency (IDF)**. It highlights words that are important in a document while down-weighting common, less informative words.

Formula For TF-IDF

$$TF(t, d) = \frac{\text{number of times } t \text{ appears in } d}{\text{total number of terms in } d}$$

$$IDF(t) = \log \frac{N}{1 + df}$$

$$TF - IDF(t, d) = TF(t, d) * IDF(t)$$

Code 🙋 X_train = tfidf.fit_transform(train_df['Review'])

X_test = tfidf.transform(test_df['Review'])

y_train = train_df['Rating']

y_test = test_df['Rating']

Output 🙋

	ability	able	absolute	absolutely	absorb	absorbed	absorbs	acai	accept	acceptable	...	zero	zest	zevia	zico	zinc	zing	zinger	zip	ziploc	ziplock
0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	...	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
1	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	...	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
2	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	...	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
3	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	...	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
4	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	...	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...
19995	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	...	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
19996	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	...	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
19997	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	...	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
19998	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	...	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
19999	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	...	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0

20000 rows × 5000 columns